

Lab Report 1: Introduction: Lenses, Telescopes and Focus/Collimation

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Experimental Setup Used: #2

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1 Executive Summary

This experiment explores the behavior of lenses and telescopic systems to understand key properties of image formation, orientation and magnification. Using optical elements such as field lens, condenser lens, and projection systems, the setups demonstrated how adjusting focal lengths between components affected the image's orientation and size. The Keplerian telescope produces inverted images approximating converging parallel ray behavior from an infinite source object. The condenser lens depended on focal lengths of 6 cm and 3.5 cm when reversed, proving geometric lens behavior. In the multi lens systems, the resulting magnifications of -0.8 and 0.8 proved theoretical predictions of focal length and image conversion behavior. Finally, adjusting the aperture sizes showed intensity and resolution correlations. At larger aperture sizes, the intensity increased at the cost of image resolution. At smaller aperture sizes, the resolution drastically improved at the cost of brightness. These results obeyed basic optical laws used in microscopes, telescopes, and other optical systems.

2 Introduction

The lab focused on optical properties such as focal length, object distance, and aperture size corresponding to image formation, magnification, and resolution. These are essential in instruments such as microscopes, telescopes, and photonics.

The first objective of the lab is to explore the nature of the thin lens equation (1):

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i} \quad (1)$$

f - focal length

d_o - the object distance

d_i - the image distance

This key equation determines the image plane formation. When the object distance is much greater than $2F$, the lens treats the object as a point source from infinity. The light at an infinite point source radiates spherical waves, approximated down to parabolic waves through fresnel approximation, and then into fraunhofer planar waves. The resulting infinite object plane's parallel rays converge to focus on the focal plane behind the lens.

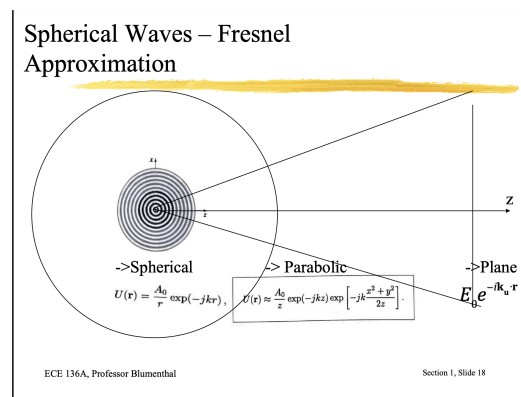


Figure 1: Approximation of Spherical Waves from ECE 136A, Blumenthal Section 1, slide. 18

1. An object is placed slightly beyond $2F$, the image produced is real, inverted and shrunk.
2. An object placed at $2F$ results in an image that is real, inverted, and equal in magnitude.
3. An object placed between $2F$ and F results in an image that is real, inverted, and enlarged
4. An object at F results in outgoing rays that are parallel (collimated light)
5. An object placed closer than F produces a virtual, upright, enlarged image

The first objective also explores the Keplerian telescope configuration. In this setup a field lens and condenser lens are positioned in focal lengths that correlate with each other to explore the nature of collimated parallel rays. The magnification of the setup is governed by the linear magnification equation (2):

$$M = \frac{h_i}{h_o} = -\frac{d_i}{d_o} \quad (2)$$

M - the magnification

h_i - the image height

h_o - the object height

d_i - the image distance (from the lens)

d_o - the object distance (from the lens)

The negative sign in the magnification equation indicates upright when positive. Here the image is virtual, indicating that it appears like it was created from virtual rays on the object side of the lens. When the magnification is negative the image is real in the back focal image plane and inverted.

The second objective explores the effect of focal length on magnification in the projection system, and the variable effects on multi-lens systems, such as the 4-f system.

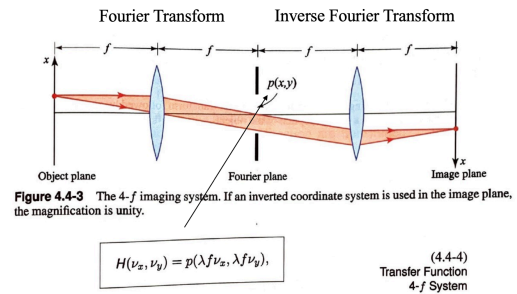


Figure 2: 4-F Imaging System from Section 3, pg. 4

The pupil imaging method is designed to form sharp images. It follows the light path that conjugates through the object and image planes of the system. In this method, the pupil function controls the range of ray angles that form images. This directly focuses image sharpness. The overall goal is to align the object and image planes through the corresponding lens pupil functions so each point correlates to a focused point in the image plane.

Single Lens Pupil Imaging Function

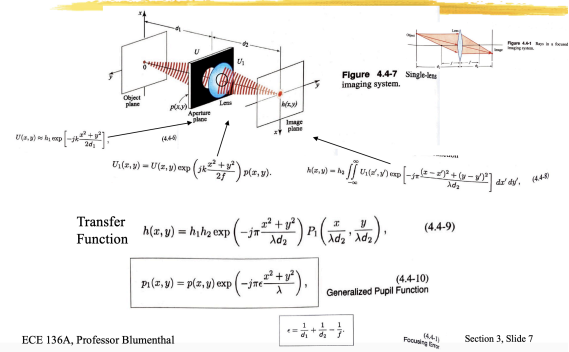


Figure 3: Pupil Imaging Function from Section 3, pg. 7

The second objective is observed through microscope magnification (3). This is the combined effect of the objective lens and eyepiece lens, which together affect the magnification through two stages in the 4F system. The objective lens forms a real, inverted image. Then the second stage magnifies the intermediate image which re-inverts the image and returns the final magnification of the system.

First stage objective lens magnification

$$V_{\text{obj}} = \frac{t}{f_{\text{obj}}}$$

Second stage eyepiece lens magnification

s_0 - reference scaling factor that converts angular eyepiece into linear magnification

$$V_{\text{EP}} = \frac{s_0}{f_{\text{EP}}}$$

Final microscope magnification

$$V_{\text{Microscope}} = \frac{t \cdot s_0}{f_{\text{obj}} \cdot f_{\text{EP}}} \quad (3)$$

from ECE 136A, Blumenthal Section 3, slide 11

The microscopic magnification equation (3) reinforces how magnification is multiplied through stages in multi lens systems.

The final objective explores the final projection of the object plane to the image plane in the 4-F system and the fourier filtering plane dictated by aperture size.

Aperture size and diffraction effects are considered in relation to resolution. From the Rayleigh criterion, two points are resolvable at the maximum of one diffraction pattern. This pattern relates to the first minimum of the other. The objective is to explore the relation of aperture diameter size on resolution and brightness. Aperture size explores the relationship between ray optics and fourier imaging principles where filtered frequencies of lights affect the resolution and brightness of images. The aperture size's diameter is determined by an optical element whose knob controls pinhole size.

Spatial Filter Pinhole

Useful for creating a plane like wave and cleaning up the spatial profile of a beam

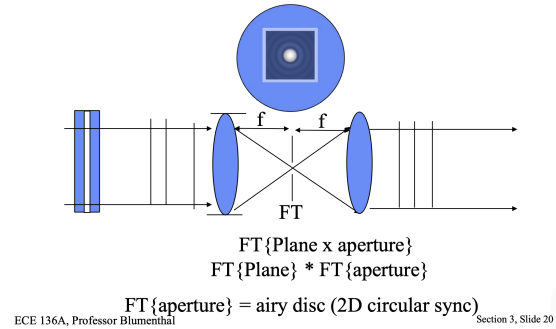


Figure 4: From ECE 136A, Blumenthal spatial Filter Pinhole section 3, slide 20

Theoretical predictions from the thin lens equation(1), telescope magnification (3), and diffraction were quantitatively verified.

These foundations help explore more complicated optical systems.

3 Experimental Procedure and Data

3.1 Collimation, Keplerian Telescope

Part 1 - Field Lens Effect on Collimated Light

For the first part of the experiment, a collimated LED beam and a field lens are aligned so that the LED beam is centered to the field lens' optical axis. A mask is placed directly in front of the LED beam as a reference to assess the resolution, magnification, and inversion. The LED beam and mask is placed on the front focus point of the field lens.

The system is set up and a screen is moved behind that optical axis of the field lens to observe the image plane. When the screen was moved along the optical axis behind the field lens, the resolution changed slightly. As the screen moved away from the point, the image resolution decreased.

We then changed the object distance corresponding to the field lens' object plane to

notice the image's resolution, magnification, and inversion to check consistency with the thin lens equation.

To confirm that the set up obeyed the thin lens equation, the object distance was varied by manipulating the distance of the field lens from the target and LED beam, and the following was noted about magnification and image focus.

1. At an object distance of 22 cm, a clear larger inverted image was formed 55 cm behind the field lens
2. As the object distance was pushed past $2F$ the image size drastically diminished and was inverted.
3. When the object distance was placed near $2F$ the image appeared equal in magnitude and inverted.
4. When the object distance was placed between F and $2F$ the image appeared inverted and magnified
5. When the object distance was placed at a distance smaller than F there was no resolvable image on the screen

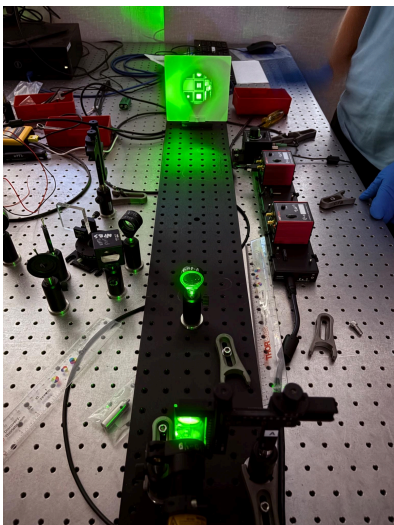


Image 1: Collimation setup showing LED laser, target mask, and screen

Part 2 - Focal Length with Condenser Lens

A condenser lens is added and aligned with the optical axis past the field lens. The screen is

now placed behind the condenser lens. The image seen on the screen appears to be a focused point of light and the best brightness appears to be observed at 6cm for the normal orientation of the condenser lens and 3cm for the flipped orientation of the condenser lens. The reference target is not resolvable and appears like a bundle of focused light.

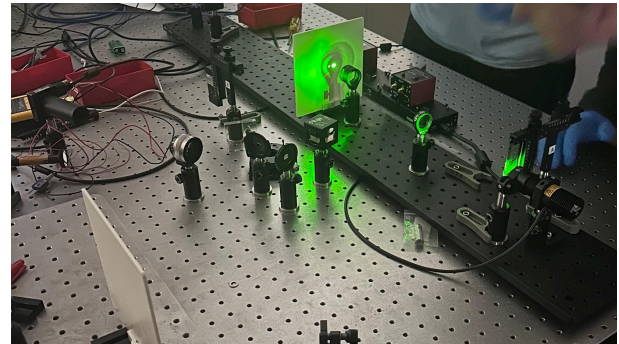


Figure 2: Condenser focusing collimated light

Part 3: Keplerian Two Lens Alignment

Following the same set up from part 2, the condenser lens was fixed at the back focus of the field lens at 15cm. The target and LED laser was fixed 15cm at the front focus of the field lens. Finally, the mask was placed at the back focus of the condenser lens.

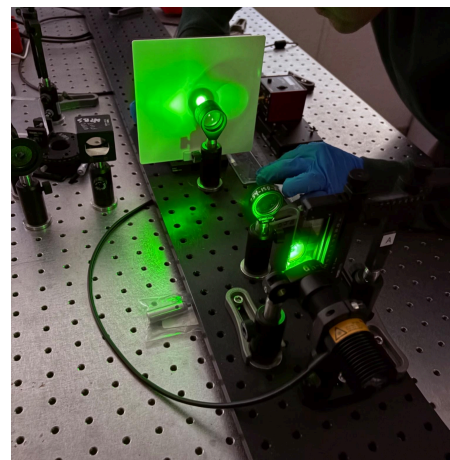


Figure 3: Keplerian lens set up

3.2 Focal Length on Magnification

The setup started with a light source placed at the front focus of a field lens (15cm). After the field lens the target was placed at the back focus of the field lens (15cm). The condenser lens, with a focal length of 5cm, was placed (5cm) after the target. Next, the objective lens, with a focal length of 3cm, was placed (3cm) behind the condenser length. Finally, a screen was placed behind the objective.

The light source was turned on and the screen was translated to find the image with best focus, which was measured to be 4cm.

The image appeared smaller than the mask and inverted.

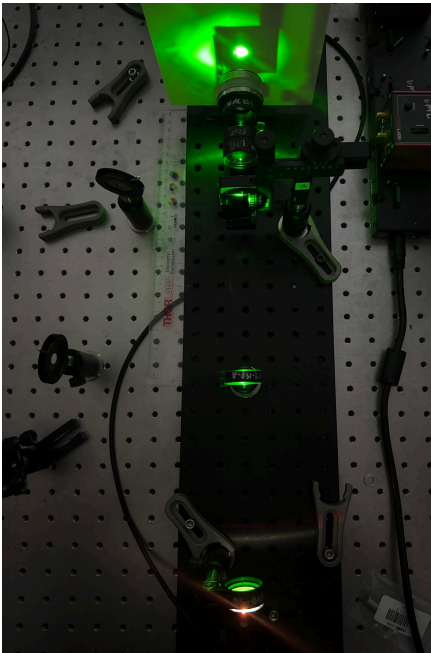


Figure 4: Optical setup for 3.2

3.3 Image Projection

Following the setup from 3.2 after the objective lens the screen was removed. Instead a projector lens, with a focal length of 7.5 cm was placed behind the objective lens. The screen was placed behind the projector lens.

1. At roughly $2F$ or 19cm away from the objective lens the object was seen as upright and the same size.
2. When the projector lens was placed at distances between 11.5cm-19cm, the image was upright and magnified.
3. When the projector was placed near the focus of the projector (around 12cm), the observed image on the screen seemed to extend past the railing of the set up, projecting onto the wall of the laboratory.
4. When the projector was placed farther than 19cm, the object was upright and seemed smaller.

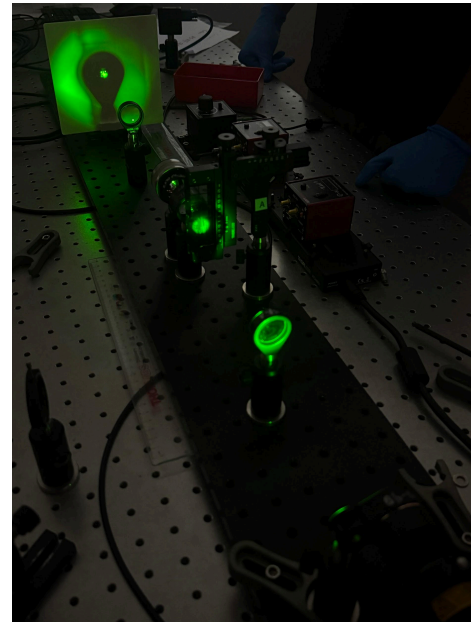


Figure 5: Optical setup for 3.3 including projector lens

Next, an adjustable pinhole aperture was placed between the condenser and objective lens. When the diameter of the aperture was set to its smallest setting the image appeared in extreme detail at a dim brightness. As the diameter of the aperture was increased, the resolution of the image decreased as the brightness increased.

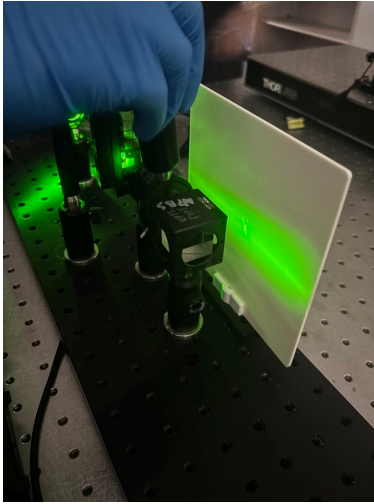


Figure 6: Small Aperture diameter resulting in fine detail and less brightness

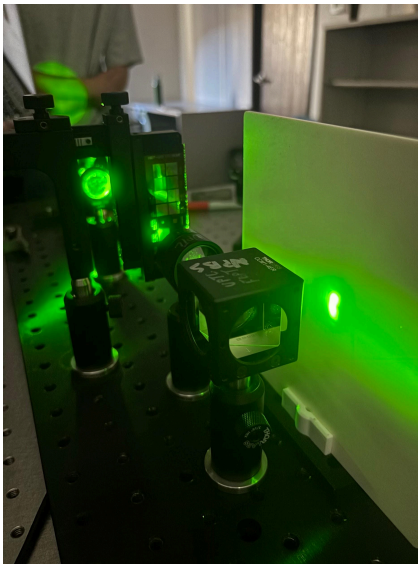


Figure 7: Larger Aperture diameter resulting in less detail and higher brightness

4 Discussion, Analysis

4.1 Collimation, Keplerian Telescope

Part 1 - Field Lens Effect on Collimated Light

This system obeyed the ray-optics model of lenses. While observing the displacement of the

screen, when the object was further than $2F$, the image appeared real, inverted, and demagnified. When the object was near $2F$, the image appeared real, inverted, and equal in size. When the object was between $2F$ and F , the image was real, inverted, and larger. At F , the light source acted as a point source and obeyed collimated light as the light source diverged to the screen.

At an object distance of 22 cm the image seemed to focus at a distance of 55 cm on the screen. The lens had a focal length of 15 cm.

Calculated the theoretical image distance obeying thin lens:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$\frac{1}{d_i} = \frac{1}{15 \text{ cm}} - \frac{1}{22 \text{ cm}}$$

$$d_i \approx 47.14 \text{ cm}$$

Theoretical Magnification:

$$M = -\frac{d_i}{d_o} = -\frac{47.14 \text{ cm}}{22 \text{ cm}} \approx -2.14$$

Experimental Magnification:

$$M = -\frac{d_i}{d_o} = -\frac{55 \text{ cm}}{22 \text{ cm}} = -2.5$$

The best measured resolution was observed at 55cm. The reason for the variance can be attributed to the human eyeball, which introduces error. What appears to be the best resolution to the human eye could actually be the wrong distance. In addition, rulers were used to check distances, and getting precise theoretical focal lengths, as well as object lengths could have been measured mistakenly

due to error. However, the theoretical and experimental data lined up because the object was inverted, appeared magnified, and real, obeying the theoretical predictions when there is a biconvex focal length between F and $2F$.

Finally, patterns did appear on the screen. This is due to the diffraction from the light source past the lens. From Kohler illumination, the light source produced interference of scattered paths, which were more visible from better focuses.

Part 2 - Focal Length with Condenser Lens

After the condenser lens was placed in the optical path, the collimated beam produced by the field lens hit the condenser lens. The collimated light was approximately parallel, which obeys an object point source appearing effectively from infinity.

Therefore thin lens equation effectively is reduced to:

$$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$$

$$f \approx d_i$$

During the experiment, the condenser lens in its front orientation produced the highest resolution light at 6 cm away from the screen, which is indicative of the parallel rays converging at the back focal length of the condenser lens.

When the lens was reversed, the light still converged to the back screen except it converged at a smaller distance of 3.5 cm. This is due to the asymmetric properties of the condenser lens. It is highly probable that the condenser lens does not have perfect spherical symmetry between its front and back convex lens; therefore, the distance from the back lens converged to a different focal point. During the

experiment, this was observable with the focus varying from 6 cm to 3.5 cm.

This behavior was not noticeable in all biconvex lenses used in our set up. It was particular to lenses such as the condenser lens because they are designed to have one side for the entry of light rays from the object plane and convergence of the entry light to a certain focal point in the back plane. If the condenser lens was a perfectly symmetric biconvex lens then flipping its orientation would give the same results for varying orientations.

Part 3: Keplerian Two Lens Alignment

It is not necessary for the condenser lens to be exactly at the field-lens focus for a sharp image. When the screen is placed at the back focal length of the condenser lens, the image was observed to be the same at varying condenser lens displacements. This is because the condenser lens is accepting collimated light from the field lens. However, the reason why the set up was in its specific configuration was to obey Keplerian telescope principles.

The field-condenser was a feature of the telescope. The field lens acted like an intermediate image, forming collimated light which the condenser received and diverged to its back focal point. This field-condenser optical setup with the collimated intermediate plane obeyed Keplerian telescope principles. It was not a microscope because a microscope uses a tube lens to form a large, virtual intermediate image for the eyepiece lens to interpret, which is completely different from the collimated light produced in the setup.

The image on the screen was inverted. This was because the setup had two-lens to make one component that produced a first inversion and a second reinverting inversion to upright.

The magnification of this telescope is the following:

$$M \approx -\frac{15}{5} = -3$$

The focal length was 15 cm and the back focal length with the screen was placed 5 cm away from the condenser. This is an approximation and could deviate because of human error with ruler measurements.

Keeping the distances between the three components (field, condenser, and screen) constant and moving the object plan before the field, placed the light source inside the focal length. Instead of producing collimated light, the field lens produced diverging rays. Because the light source is a bundle of rays, the diverging rays still formed an image that was defocused and appeared larger. This obeys ray optics because the image distance is constant and the object distance is reduced, increasing the magnitude of magnification on the image.

In order to replicate the previous image, the object distance needed to return back to the focal length of the field lens or the screen needed to be adjusted to the new theoretical image plane. This obeyed what was observed in the experiment when the screen was moved to reorient the image produced from the three element shift.

4.2 Focal Length on Magnification

The best observable focus was found to be 4 cm behind the objective where the image was seen to be inverted. This set up introduced a light source at the front focal length of the field lens, resulting in collimated light. The target placed at the back focal length of the field lens acted as a backlight object that was placed at the front focal of the condenser lens. The back light object at the front of the focal condenser lens produced collimated light once again, and then was converged to the back focal length of the objective lens.

Theoretically, the objective lens had a focal length of 3cm and should have produced an

inverted, real image with magnification $-\frac{3}{5}$. However, due to human error and the bulkiness of the objective lens, there was deviation from the true theoretical principal planes. The image focused best at 4cm behind the objective lens and was found to have a magnification of $-\frac{4}{5} = -0.8$. Additionally, the backlight object could deviate slightly and not act perfectly as a point source, so the best focus visualization on the screen could produce experimental uncertainty.

This setup was different from the previous setup, particularly in magnification. Instead of having two lens work and one column of collimated light to produce an inversion, there was two sets of collimated light and the inversion was affected by the distance from the backlit mask (acting as a source object) then passed through the condenser lens as the second wave of collimated light to get condensed onto to the final screen. This affected the distance of the object in the magnification. Instead of it being from the first focal length from the light source to the field lens in the first setup, now the distance of the object was the focal length of the back light object to the condenser lens.

4.3 Image Projection

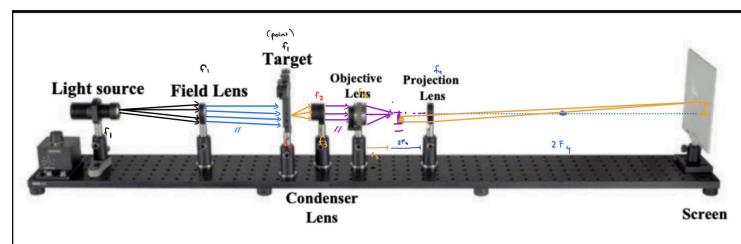


Figure 8: Ray optics diagram optical set up

The objective lens formed a real, inverted image with a magnification found to be -0.8, varying slightly from the theoretical prediction of -0.6. This image acted as an intermediate image at the back focal point of the objective lens. A projector lens was oriented so that the image plane from the back focal length of the objective is $2F$ (15cm) away from the projector lens or a total of (4cm+15cm=19cm away).

Theoretically the projection onto the screen on the back of the projector lens should focus at 2F (15cm) from the back of the projector lens and should be inverted for the second time, making it upright.

The magnification of the final image was calculated to be:

Objective magnification: -0.8

At the object distance near 2F:

$$\frac{1}{7.5} = \frac{1}{15} + \frac{1}{15}$$

$$M \approx -\frac{15}{15} = -1$$

$$M_{\text{total}} = M_{\text{obj}} \times M_{\text{proj}} \approx (-0.8) \times (-1) = +0.8$$

The final image from the optical setup obeyed theoretical predictions with experimental variations and was upright and real (+0.8).

The plane past the condenser before the objective lens is the Fourier plane. The collimated light produced in between the condenser and objective lens consists of plane waves with distinct spatial frequencies. This is called the Fourier plane because the plane is subject to Fourier transforms in the form of apertures.

When an aperture was placed between the Fourier plane of the condenser and objective lens the image appeared bright and blurry versus dim and clear in resolution. This showed the aperture working the Fourier transform in the Fourier plane, filtering light frequencies.

When the aperture was at larger diameters it worked as a high pass filter and lets higher spatial frequencies through which produced with brighter light on the screen but more blurring. When the aperture diameter was diminished it acted as a low pass filter, filtering

larger spatial plane waves and only letting low frequencies through. This produced a higher resolution up to a certain diffraction limit at the expense of dimmer light objects on the screen. This can be explained through the Rayleigh criterion for a circular pupil (the aperture diameter) where the central maximum of one diffraction pattern aligned with the first minimum of the other pattern, affecting image resolution. Rayleigh criterion applications in industry can be seen everywhere from optical microscopes and other usages where inspecting tiny details is required such as wafer inspection.

5 Conclusion

The first part of the lab proved the thin lens equation through collection of object and image distances, magnification, and orientations. The field lens in this set up produced an inverted image that varied from object placement. In addition, the setup verified the convergence of collimated light produced from the field lens. These principles are seen in Keplerian telescope configurations.

The second part of the lab focused on multi-lens systems and showed how two sources of collimated light can vary magnification. Adding the projector lens reoriented the initial inversion to an upright nature and showed how magnifications multiply in multi-component optical setups. Finally, introducing apertures between Fourier planes showed how diffraction limits play a role in resolution and brightness. Overall, the work and data collected in this lab proved theoretical relationships in basic optical setups and set the foundation to understand microscopes and telescope optics.

References

1. Saleh, B. E. A., & Teich, M. C. *Fundamentals of Photonics*, 2nd ed., Wiley, 2007.
2. Lecture Slides, "Week 1-3" Prof. Blumenthal, UCSB ECE 136A, Fall 2025.